

Unit 6

SECURITY, PRIVACY AND DATA INTEGRITY

32 items · 163 marks

Syllabus sections in this unit

6.1 Data Security

6.2 Data Integrity

Unit 6: Security, privacy and data integrity

1. M/J 26/P11/Q4/d

- 4 A shop allows customers to pay to borrow toys. The shop uses a database to record information about the customers and the toys they have borrowed.

CUSTOMER (CustomerID, ContactFirstName, Phone)

TOY (ToyID, ToyName, CostPerDay)

TOY_INSTANCE (UniqueToyID, ToyID, Quality)

BORROW (BorrowID, CustomerID, UniqueToyID, StartDate, NumberDays)

- (d) When data is entered into the database it is validated and verified.

- (i) Explain how validation and verification support data integrity within a database.

[4]

- (ii) Give **one** reason why data may still be incorrect even after it passes validation and verification.

[1]

2. M/J 26/P12/Q2/a,b,c

- 2 Personal data is entered into a computer using a keyboard. The data is validated and verified during data entry.

- (a) Describe **two** methods of data verification that can be used during data entry.

1

2

[4]

- (b) The personal data includes the date of birth, for example 05/07/2010.

Complete the table by identifying **two** methods of validation that can be used on the date of birth.

State how each validation method will be used for the date of birth.

data validation method	use
.....	
.....	

[3]

(c) The data needs to be kept secure whilst on the computer.

(i) A virus is a threat to data.

Complete the table by describing what is meant by a virus.

Identify **and** describe **one** method to restrict the risk posed by a virus.

virus description	
method to restrict risk	
description of method to restrict risk	

[3]

(ii) A hacker is another threat to data.

Complete the table by describing what is meant by a hacker.

Identify **and** describe **one** method to restrict the risk posed by a hacker.

hacker description	
method to restrict risk	
description of method to restrict risk	

[3]

3. M/J 26/P13/Q1

- 1** Validation and verification can be used to protect the integrity of data.

Complete the description of methods of ensuring data integrity by writing the missing terms.

..... is a method of verification where the data is entered by a user and then manually compared to the original data to make sure that they are the same.

A verification method used during data transfer calculates a value from the data which is transferred with the data. The receiver performs the same calculation and compares the result with the value transferred.

This method of verification is

There are several methods of data validation.

check compares the data entered to a list of possible values that can be entered. If the data is not one of these values it is rejected.

..... check can be used to make sure data entered has the correct combination of letters, numbers and symbols. For example, a code may need to be two capital letters, one number followed by one symbol.

..... check is used to make sure the correct quantity of characters has been entered. For example, a name must be more than one character and less than or equal to 100 characters.

[5]

4. O/N 25/P11/Q8/a

- 8** One ethical consideration for a student connecting their personal computer to the school network is the risk of spreading malware on the network.

(a) Viruses and pharming are examples of malware.

Explain what is meant by a virus and pharming.

Virus

Pharming

[2]

5. O/N 25/P12/Q1

- 1** Draw **one** line from each verification method to indicate whether it is used during data transfer or data entry.

Verification Method	Use
Parity byte check	
Checksum	Data transfer
Visual check	
Parity block check	Data entry

[2]

6. O/N 25/P12/Q5/e

- 5** A local area network (LAN) has five computers, one switch and one server.

- (e)** Complete the table by identifying **one** threat to computer and data security posed by networks and the internet.

Describe the threat and give a method of prevention.

Threat	Description	Prevention

[3]

7. O/N 25/P12/Q6/c

- (c) A computer system uses even parity. The least significant (rightmost) bit of each byte is the parity bit.

- (i) Complete the byte by writing the missing parity bit:

							parity bit ↓
0	1	0	1	1	1	0	

[1]

- (ii) The computer also uses parity block check. The parity block check uses even parity. Computer A transmits four bytes of data to computer B, followed by a parity byte. Computer B receives the following sequence of bytes.

							parity bit ↓
1	0	1	1	0	1	1	1
0	1	1	1	0	0	0	0
0	0	0	1	1	0	1	1
0	1	1	1	0	1	0	0
parity byte →	1	0	1	0	0	0	0

Following transmission, one of the four bytes of data has an error in one of the bits.

Circle the bit that has been altered during the data transfer.

[1]

8. O/N 25/P12/Q10/c

- 10 A programmer is developing a computer program.

- (c) A user downloads the computer program from the internet.

State what should be included as part of the download to make sure the program is authentic.

[1]

9. O/N 25/P13/Q7/c,e

- 7 A digital video camera records students in a classroom. The data is transferred from the digital video camera over the internet to a server. Artificial Intelligence (AI) is used to analyse the video on the server to identify when students are interacting with the lesson and the teacher.

- (c) Identify **and** describe **one** method of data verification that can be used when transferring data from the digital video camera to the server.

Method

Description

[3]

- (e) The table contains three algorithms that perform data validation. The third algorithm uses the functions MID and LENGTH.

The functions MID and LENGTH are defined as follows:

- `MID(ThisString : STRING, x : INTEGER, y : INTEGER)` RETURNS STRING
returns a string of length `y` starting at position `x` from `ThisString`
- `LENGTH(ThisString : STRING)` RETURNS INTEGER
returns the integer value representing the length of `ThisString`

Complete the table to identify the method of data validation for each algorithm. Each method of data validation must be different.

Algorithm	Method of data validation
<pre> INPUT x IF x < 1 or x > 26 THEN OUTPUT "Invalid" ENDIF </pre>	
<pre> INPUT x IF x <> 'R' AND x <> 'G' AND x <> 'B' THEN OUTPUT "Invalid" ENDIF </pre>	
<pre> INPUT x FLAG ← FALSE FOR INDEX ← 1 TO LENGTH(x) IF MID(x, INDEX, 1) = "@" THEN FLAG ← TRUE ENDIF NEXT INDEX IF NOT FLAG THEN OUTPUT "Invalid" ENDIF </pre>	

10. O/N 25/P21/Q6/a

- 6 Students are learning about a simple check digit method for data validation. In this method, a single check digit is appended to the end of an original number to give a new number.

The students are studying a method which:

- calculates the sum of all the digits in the original number
- uses integer division to calculate the remainder when the sum is divided by 10
- uses the remainder as the check digit
- appends the check digit to the original number, creating the new number.

For example:

original number	4162
sum of all digits	$4 + 1 + 6 + 2 = 13$
remainder when the sum is divided by 10 using integer division	3
new number	41623

The method described can be used to detect single-digit errors. For example, if the new number is incorrectly input as 41633, then the check digit does not match and the input will be rejected.

- (a) An incorrect attempt was made to enter 41623. The **first two** digits were entered incorrectly; the **last three** digits were entered correctly.

When this number was tested, the check digit was found to be correct and the input was accepted.

Identify an example of the incorrect attempt to enter 41623 and explain why this number would **not** be rejected.

Number

Explanation

[2]

11. M/J 25/P11/Q7/a,b,c

7 A banker stores personal data on their work computer.

(a) The banker needs to transfer confidential data across the internet.

Identify and describe **one** method of restricting the risks posed by an unauthorised person intercepting the data whilst it is being transferred across the internet.

Method

Description

[3]

(b) The banker receives confidential data across the internet. The data includes a digital signature.

Explain how a digital signature can make sure the data has **not** been changed during transmission.

[5]

(c) The data that is transferred can also be verified using a checksum.

Explain how data can be verified using a checksum.

[3]

12. M/J 25/P13/Q3/b

3 A programmer is writing a program in a high-level language.

- (b)** The programmer needs to keep the program files secure on their computer and during electronic data transmission over the internet. The files are protected by a password.

Complete the table by identifying **one** other method of keeping the files secure during electronic data transmission **and one** other method of keeping the files secure on the computer.

State how each method protects the data.

The methods must be different.

	Method	How the method protects the data
during transmission		
on the computer		

[4]

13. O/N 24/P12/Q4/c

4 A program is written in a high-level language by a team of three programmers using an Integrated Development Environment (IDE).

- (c)** The file containing the final program code will be sent by email for beta testing.

Identify **one** security method that can be used to protect the program code from unauthorised access during email transfer.

Explain how your chosen method protects the program code.

Security method

Explanation

[3]

14. M/J 24/P11/Q5/b,c(i)

- 5** A bank allows customers to access their accounts using an application that they can download onto a device such as a smartphone.

- (b)** The bank wants to protect the integrity of its data while transferring the data to other banks. Parity check is one example of data verification.

Complete the description of parity check when Computer A is transmitting data to Computer B.

Computer A and Computer B agree on whether to use

parity. Computer A divides the data into groups of The

number of 1s in each group is counted. If the agreed parity is

and the group has an even number of 1s, a parity bit of 1 is appended, otherwise a parity bit of 0 is appended.

In a parity check the bytes are grouped together, for example in a grid. The number of 1s in each column (bit position) is counted. A bit is assigned to each column to make the column match the parity. These parity bits are transmitted with the data as a parity

[5]

- (c)** The bank also needs to keep its customers' data private and secure.

- (i)** The bank's network has a firewall.

Explain how a firewall can help protect the customers' data.

[3]

15. M/J 24/P12/Q3/a

- 3** An assessment board scans exam papers and stores the digitised papers on a server. Exam markers download the digitised papers to mark. The exam markers then upload the mark for each paper.

(a) The assessment board needs to make sure the data stored on the server is secure.

- (i)** Authentication methods can help to protect the server against hackers.

Identify **one other** security measure that helps to protect the server from hackers.

Describe how the security measure works.

Security measure

Description

[3]

- (ii)** Identify **one** security measure that helps to protect the data when it is being transmitted to its destination. Describe how the security measure works.

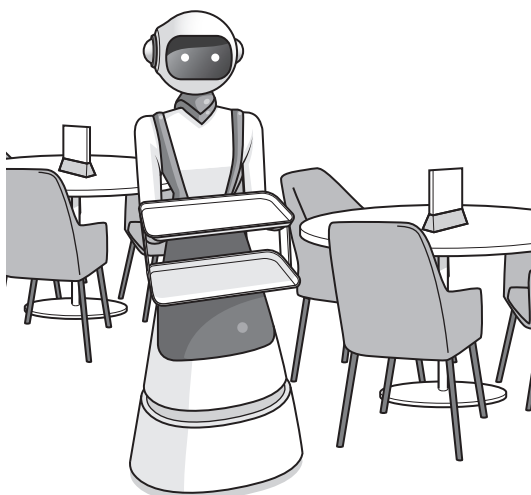
Security measure

Description

[3]

16. M/J 24/P13/Q7/f

- 7** Robots are used to serve food and drink to customers at a restaurant.



(f) The data from the robots is transmitted to a central computer using a wireless connection.

- (i)** Complete the table by identifying **and** describing **two** methods of data verification that can be used during data transfer.

	Method	Description
1		
2		

[4]

(ii) Explain how encryption can protect the security of data during transmission.

[2]

17. O/N 23/P12/Q5/a,b,c

5 (a) State the meaning of **privacy of data**.

[1]

(b) State the meaning of **integrity of data**.

[1]

(c) Describe the following threats to a computer system.

Phishing email

Spyware

[4]

18. O/N 23/P13/Q3/a(iii),c

- 3** A company sells online Computer Science courses to students in different countries.

The courses are stored on a public cloud.

- (iii)** State how the following security measures can be used to protect computer systems.

Firewall

Encryption

Passwords

[3]

- (b)** The company uses a database, `COURSES`, to store data about the courses and their tutors.

Each course starts at different times of the year and may have a different tutor.

The database has the following structure:

`COURSE_INFORMATION(CourseID, Description, Cost)`

`TUTOR(TutorID, TelephoneNumber, EmailAddress, TutorName)`

`COURSE_SCHEDULE(CourseID, DateStarted, TutorID)`

- (c)** An example of a tutor ID is NK16C6.

An administrative officer enters the tutor ID into the `TUTOR` table.

Explain how data verification can be used when the tutor ID is entered.

[4]

19. O/N 23/P13/Q8/a,b,c

- 8** **(a)** Data verification is one method of protecting the integrity of data.

Describe **one** other method of protecting the integrity of data.

[2]

- (b)** State **one** difference and **one** similarity between pharming and phishing.

Difference

Similarity

[2]

- (c)** Explain how the data security risks of malware can be restricted.

[3]

20. M/J 23/P12/Q2/c(i)

- 2** A horse riding school uses a database, `Lessons`, to store data about lesson bookings.

This database is created and managed using a Database Management System (DBMS).

- (c)** The database `Lessons` has the following tables:

`HORSE(HorseID, Name, Height, Age, HorseLevel)`

`STUDENT(StudentID, FirstName, LastName, RiderLevel, PreferredHorseID)`

`LESSON(LessonID, Date, Time, StudentID, HorseID, LessonContent)`

Dates in this database are stored in the format `#DD/MM/YYYY#`.

The fields `RiderLevel` and `HorseLevel` can only have the values: `Beginner`, `Intermediate` or `Advanced`.

- (i)** Describe **two** methods of validating the field `RiderLevel`.

1

2

[2]

21. M/J 23/P13/Q4/d

- 4** A shop rents cars to customers. The shop uses a relational database to store information about the rentals.

(d) The data is validated and verified when it is entered into the database.

- (i)** The car registration number must be: 1 letter, followed by 3 numbers, followed by 2 letters.

For example, A123AA is valid but A12AA is invalid.

One way that a registration number can be validated is by using a presence check to make sure the registration number has been entered.

Describe **two other** ways that the car registration number can be validated.

1

2

[2]

- (ii)** Describe **two** ways that the car registration number can be verified when it is entered into the database.

1

2

[2]

- (iii)** State why the car registration number might be incorrect even after it has been validated and verified.

[1]

22. M/J 23/P13/Q6/a,b

6 Data needs to be kept secure when stored on a computer and during transmission over a network.

- (a) Explain how a digital signature is used to authenticate a digital document during transmission over a network.

[5]

- (b) Complete the table by identifying **and** describing **two** types of software that can be installed on a computer to prevent threats over a network.

Type of software	Description
.....	
.....	

[2]

23. M/J 23/P23/Q2/b

2 A program stores a date of birth for a student using a variable, `MyDOB`, of type `DATE`.

- (b) A function `CheckDate()` will take three integer parameters representing a day, month and year of a given date.

The function will validate the date of birth for a student that the parameters passed to it represent.

For a date to be valid, a student must be at least 18 in year 2020.

- (i) Two of the parameter values can be checked without reference to the third parameter.

Describe these **two** checks.

Check 1

Check 2

[2]

- (ii) Several values of the parameter representing the day can only be checked completely by referring to the value of **one other** parameter.

Describe this check.

[2]

24. O/N 22/P11/Q2

2 Draw **one** line from each security feature to its most appropriate description.

Security feature	Description
firewall	converts data to an alternative form
pharming	redirects a user to a fake website
anti-virus software	verifies the authenticity of data
encryption	scans files on the hard drive for malicious software
	accepts or rejects incoming and outgoing packets based on criteria

[4]

25. O/N 22/P12/Q4/a,b,c

4 (a) State the difference between **data verification** and **data validation**.

[1]

(b) A checksum can be used to detect errors during data transmission.

Describe how a checksum is used.

[3]

(c) One validation method is a presence check.

Describe **two other** validation methods that can be used to validate non-numeric data.

1

2

[2]

26. O/N 22/P12/Q6/a

6 (a) A student uses a networked laptop computer to send an email to a colleague.

(i) Explain how a digital signature ensures the email is authentic.

[2]

(ii) Describe how a firewall protects the data on the computer.

[3]

27. O/N 21/P11/Q2/a,b,c

2 Xanthe wants to maintain the integrity and security of data stored on her computer.

(a) Explain the difference between data security and data integrity.

[2]

(b) Xanthe uses both data validation and data verification when entering data on her computer.

(i) Describe how data validation helps to protect the integrity of the data. Give an example in your answer.

Description

Example

[2]

(ii) Describe how data verification helps to protect the integrity of the data. Give an example in your answer.

Description

Example

[2]

(c) Two malware threats are spyware and viruses.

Give **two** similarities and **one** difference between spyware and a virus.

Similarity 1

Similarity 2

Difference

[3]

28. O/N 21/P12/Q1

- 1 When designing computer systems, it is important to consider the security, integrity and privacy of the data.

Draw **one** line from each measure to indicate whether it keeps data secure or protects the integrity of data.

Measure

Firewall

Double entry

Presence check

Access rights

Password

Data Security

Data Integrity

[2]

29. O/N 21/P13/Q2/a,b,c

2 Xanthe wants to maintain the integrity and security of data stored on her computer.

(a) Explain the difference between data security and data integrity.

[2]

(b) Xanthe uses both data validation and data verification when entering data on her computer.

(i) Describe how data validation helps to protect the integrity of the data. Give an example in your answer.

Description

Example

[2]

(ii) Describe how data verification helps to protect the integrity of the data. Give an example in your answer.

Description

Example

[2]

(c) Two malware threats are spyware and viruses.

Give **two** similarities and **one** difference between spyware and a virus.

Similarity 1

Similarity 2

Difference

[3]

30. M/J 21/P11/Q6/a,b,c

6 Each of the following algorithms performs data validation.

State the type of validation check that each of the algorithms performs.

(a)

```
INPUT x  
  
IF x < 0 OR x > 10 THEN  
  
    OUTPUT "Invalid"  
  
ENDIF
```

[1]

(b)

```
INPUT x  
  
IF x = "" THEN  
  
    OUTPUT "Invalid"  
  
ENDIF
```

[1]

(c)

```
INPUT x  
  
IF NOT(x = "Red" OR x = "Yellow" OR x = "Blue") THEN  
  
    OUTPUT "Invalid"  
  
ENDIF
```

[1]

31. M/J 21/P12/Q8/a,b,c

- 8** A company has several security measures in place to prevent unauthorised access to the data on its computers.

(a) Describe the difference between the security and privacy of data.

[2]

- (b)** Each employee has a username and password to allow them to log onto a computer. An employee's access rights to the data on the computers is set to either read-only, or read and write.

Identify **one** other software-based measure that could be used to restrict the access to the data on the computers.

[1]

- (c)** The company is also concerned about threats posed by networks and the internet.

Identify **two** threats to the data that are posed by networks and the internet.

Threat 1

Threat 2

[2]

32. M/J 21/P13/Q6/a,b,c

6 Each of the following algorithms performs data validation.

State the type of validation check that each of the algorithms performs.

(a)

```
INPUT x  
  
IF x < 0 OR x > 10 THEN  
  
    OUTPUT "Invalid"  
  
ENDIF
```

[1]

(b)

```
INPUT x  
  
IF x = "" THEN  
  
    OUTPUT "Invalid"  
  
ENDIF
```

[1]

(c)

```
INPUT x  
  
IF NOT(x = "Red" OR x = "Yellow" OR x = "Blue") THEN  
  
    OUTPUT "Invalid"  
  
ENDIF
```

[1]

Answers Section

1. M/J 26/P11/Q4/d

4(d)(i)	1 mark for each correct point to max 4 - Validation only ensures that data meets certain criteria - Validation prevents the input of inappropriate data - Verification makes sure that the data that is entered matches the original data / the data that was intended to be entered - Verification prevents data being accidentally changed from the original - Any example of validation/verification in practice e.g. entering a password twice	4	Accept a type of validation for MP1 BOD or a type of verification for MP3 BOD but not for both
4(d)(ii)	1 mark for a correct answer The original data was incorrect // the same wrong data was entered twice, and the wrong data was also valid // by example	1	User error is TV

2. M/J 26/P12/Q2/a,b,c

2(a)	1 mark for method 1 mark for correct matching description (* 2) - Visual check - The entered data is compared with the original by a person - Double entry - Enter the data twice and the computer automatically compares if they match	4	Note: Question asks for methods of data verification not validation Accept examples of visual checks, such as printing a copy and comparing the printout to the original Data as intended is NE. It is the comparison to the original that is important
2(b)	1 mark for two correct names of methods 1 mark for each correct matching use to max 2 - Presence check - The Date of Birth (DoB) field cannot be left blank - Range check - DoB must be after one date and before another date // between specific dates – must include both ends - Limit check - DoB must be after or before a specific date – must be only one end - Format check - DoB must be dd/mm/yyyy - Length check - DoB must be exactly 10 characters - Type check - The data type must be DATE or DATE/TIME	3	Note: Question asks for methods of data validation not verification Mark the first answer in each box The use must match the example given in the question 05/07/2010 Do not accept range and limit as two different checks Take care that answers do not just repeat the method, must be applied in some way. The use must match the method If no methods are given read the uses which must be applied to the DoB. For format check accept mm/dd/yyyy
2(c)(i)	1 mark for correct description of a virus Description: malicious software that replicates on the computer and can fill storage or delete/corrupt/change data - cause harm is NE 1 mark each for a correct method and matching description, 2 marks Method: Anti-virus software // anti-malware software Description: The software scans files/downloads/device and compares against database of known viruses, and quarantines/deletes/alerts the user if a match found	3	The description of the method must match the method chosen

2(c)(ii)	1 mark for correct description of a hacker Description: person who gains unauthorised access to a computer system 1 mark each for a correct method and matching description, 2 marks For example: Method: Firewall Description: The firewall monitors all incoming and outgoing data compares it to stored criteria and blocks transmissions that are not allowed Method: Multi-factor authentication Description: A second form of verification such as biometrics or use of an authenticator app is required for logging in Method: <u>Strong</u> passwords Description: - The password contains a mixture of different types of characters and is at least a set number of characters and is not something connected to the individual such as a date of birth Method: Encryption Description: Scramble/encode the data/turn into ciphertext so that even if it is accessed it will not be understood/useable without the decryption key	3 The description of the method must match the method chosen Just malicious intent is NE for unauthorised access Accept two-factor for multi-factor If firewall is the candidate's answer for question 2(c)(i) ensure it is not repeated here Just authentication is NE for multi-factor authentication Map biometrics/use of biometric password as the method to strong password Description: Uses features of an individual that are unique and which are almost impossible to duplicate
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3. M/J 26/P13/Q1

1	1 mark for each correct term Visual check is a method of verification where the data is entered by a user and then manually compared to the original data to make sure that they are the same. A verification method used during data transfer calculates a value from the data which is transferred with the data. The receiver performs the same calculation and compares the result with the value transferred. This method of verification is checksum. There are several methods of data validation. Existence check compares the data entered to a list of possible values that can be entered. If the data is not one of these values, it is rejected. Format check can be used to make sure data entered has the correct combination of letters, numbers and symbols. For example, a code may need to be two capital letters, one number followed by one symbol. Length check is used to make sure the correct quantity of characters has been entered. For example, a name must be more than one character and less than or equal to 100 characters.	5
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4. O/N 25/P11/Q8/a

8(a)	1 mark for each correct explanation, max 2 marks Virus: Malicious program/software that replicates/copies itself and deletes/alters files/data stored on a computer Pharming: Malicious code/software installed on a computer which redirects user to a fake website to obtain personal data	2
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5. O/N 25/P12/Q1

1	1 mark for 2 or 3 correctly connected verification method boxes, 2 marks for all 4 correct Verification Method Parity byte check Checksum Visual check Parity block check Use Data transfer Data entry	2
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6. O/N 25/P12/Q5/e

5(e)	1 mark for correct threat 1 mark corresponding description 1 mark for correct method of prevention <table border="1"> <thead> <tr> <th>Threat</th><th>Description</th><th>Prevention</th></tr> </thead> <tbody> <tr> <td>Malware / Virus</td><td>Malicious code that can alter/delete files</td><td>Anti-virus // Anti-malware // Firewall</td></tr> <tr> <td>Spyware</td><td>Records keystrokes which are sent to third party</td><td>Anti-spyware // Firewall // Anti-malware</td></tr> <tr> <td>Hacking</td><td>Gaining <u>unauthorised</u> access to a computer network / device</td><td>Authentication // Firewall</td></tr> <tr> <td>Phishing</td><td><u>Emails</u> supposedly from reputable companies are sent to trick people into revealing personal information</td><td>Use a spam filter // Do not open emails from unknown sources</td></tr> <tr> <td>Pharming</td><td>Users are directed to a bogus <u>website</u> that looks legitimate to obtain personal information</td><td>Use a VPN // Anti-malware// Do not open links or download attachments</td></tr> </tbody> </table>	Threat	Description	Prevention	Malware / Virus	Malicious code that can alter/delete files	Anti-virus // Anti-malware // Firewall	Spyware	Records keystrokes which are sent to third party	Anti-spyware // Firewall // Anti-malware	Hacking	Gaining <u>unauthorised</u> access to a computer network / device	Authentication // Firewall	Phishing	<u>Emails</u> supposedly from reputable companies are sent to trick people into revealing personal information	Use a spam filter // Do not open emails from unknown sources	Pharming	Users are directed to a bogus <u>website</u> that looks legitimate to obtain personal information	Use a VPN // Anti-malware// Do not open links or download attachments	3
Threat	Description	Prevention																		
Malware / Virus	Malicious code that can alter/delete files	Anti-virus // Anti-malware // Firewall																		
Spyware	Records keystrokes which are sent to third party	Anti-spyware // Firewall // Anti-malware																		
Hacking	Gaining <u>unauthorised</u> access to a computer network / device	Authentication // Firewall																		
Phishing	<u>Emails</u> supposedly from reputable companies are sent to trick people into revealing personal information	Use a spam filter // Do not open emails from unknown sources																		
Pharming	Users are directed to a bogus <u>website</u> that looks legitimate to obtain personal information	Use a VPN // Anti-malware// Do not open links or download attachments																		

7. O/N 25/P12/Q6/c

6(c)(i)	1 mark for correct answer parity bit ↓ <table border="1" style="margin-left: auto; margin-right: auto;"><tr><td>0</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td></tr></table>	0	1	0	1	1	1	0	0	1
0	1	0	1	1	1	0	0			

6(c)(ii)	1 mark for correct answer parity bit ↓ <table border="1" style="margin-left: auto; margin-right: auto;"><tr><td></td><td>1</td><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td></tr><tr><td></td><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td></td><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td><td>1</td></tr><tr><td></td><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td></tr><tr><td>parity byte→</td><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr></table>		1	0	1	1	0	1	1	1		0	1	1	1	0	0	0	0		0	0	0	1	1	0	1	1		0	1	1	1	0	1	0	0	parity byte→	1	0	1	0	0	0	0	0	[1] 1
	1	0	1	1	0	1	1	1																																							
	0	1	1	1	0	0	0	0																																							
	0	0	0	1	1	0	1	1																																							
	0	1	1	1	0	1	0	0																																							
parity byte→	1	0	1	0	0	0	0	0																																							

8. O/N 25/P12/Q10/c

10(c)	1 mark for a correct answer • Digital Signature	1
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9. O/N 25/P13/Q7/c,e

7(c)	1 mark for the name 2 marks for the matching description, max 3 marks - Parity Byte Check - a parity bit is added to each byte to make the number of 1s match the parity, odd or even - each byte can be checked on receipt and request to be resent if the byte does not match parity - Parity Block Check - a bit is added to each byte, but a parity byte is also set for each block - the location of an error can be found using vertical and horizontal parity - Checksum - A calculation is made from the data and transmitted with the data - The receiver performs the same calculation and compares with received checksum to see if they match	3
7(e)	1 mark for each validation check correctly identified, max 3 marks Range Check Existence Check Format Check	3

10. O/N 25/P21/Q6/a

6(a)	Number: Any number where the first two digits sum to 5 / 15 and the last three digits are 623 Explanation: The remainder is the same / 3 // The sum of the (first four) digits is the same / 13	2
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11. M/J 25/P11/Q7/a,b,c

7(a)	1 mark for a correct method: 1 mark each to max 2 for a corresponding description Method: Encryption Description: - Data is encoded/scrambled using a key to create cipher text - if intercepted it cannot be understood - without being decrypted using a key	3
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7(b)	1 mark each to max 5 • The sender hashes the document / message • to produce a digest • The sender encrypts the digest to create the digital signature • The message and the signature are sent to the banker / receiver • The receiver decrypts the signature to reproduce the digest • The receiver uses the same hashing algorithm on the document received to produce a second digest • The receiver compares this digest with the one from the digital signature • If both of the receiver's digests are the same the document has not changed	5
7(c)	1 mark each to max 3 • The data is put through an algorithm to create a checksum value • The data and checksum are sent to the receiver • The receiver performs the same algorithm on the data • if both checksums match the data is verified	3

12. M/J 25/P13/Q3/b

3(b)	1 mark for method and 1 mark for corresponding description During transmission e.g. • Encryption // by example such as VPN • Jumble / encode data so it cannot be decrypted/understood without the key On computer e.g. • Firewall / proxy • Filter incoming transmissions and stop any that could be attempting unauthorised access • Anti-malware • Find and delete or quarantine any malware that could delete the data / files • Encryption • Jumble / encode data so it cannot be decrypted / understood without the key • Physical method // by example • For example, the computer storing the data cannot be accessed without the key to the room	4
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13. O/N 24/P12/Q4/c

4(c)	1 mark for the security method. 2 marks for explanation Security method: Encryption Explanation • File contents are converted to cipher text • If intercepted the data cannot be understood without the decryption key	3
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14. M/J 24/P11/Q5/b,c(i)

5(b)	1 mark for each correctly completed term: • odd or even • 7-bits • odd • block • byte Computer A and Computer B agree on whether to use odd or even parity. Computer A divides the data into groups of 7-bits. The number of 1s in each group is counted. If the agreed parity is odd and the group has an even number of 1s a parity bit of 1 is appended, otherwise a parity bit of 0 is appended. In a parity block check the bytes are grouped together, for example in a grid. The number of 1s in each column (bit position) is counted. A bit is assigned to each column to make the column match the parity. These parity bits are transmitted with the data as a parity byte.	5
5(c)(i)	1 mark each to max 3: • Compares all incoming and outgoing transmissions • ... against set criteria/whitelist/blacklist • Blocks all transmissions that do not meet rules • Blocks data entering from specific ports • Blocks unauthorised/unknown internal software transmitting data	3

15. M/J 24/P12/Q3/a

3(a)(i)	1 mark for security measure 1 mark each to max 2 for how the chosen measure works: • Firewall • Checks incoming connections • ... against criteria • Blocks data from entering specific ports • Blocks data that does not meet whitelist that meets blacklist • Proxy server • Prevents devices accessing the web server directly • Intercepts any requests • Forwards the request using its own IP address • Screens returning data before sending it to the user	3
3(a)(ii)	1 mark for security measure 1 mark each to max 2 for description of the chosen measure: • Encryption • Encodes/scrambles data • ... so if it is intercepted it cannot be understood • Algorithm/key is required to decode the data	3

16. M/J 24/P13/Q7/f

7(f)(i)	1 mark for each correct method and 1 mark for corresponding description to max 4: <table><tr><th>Method</th><th>Description</th></tr><tr><td>Parity byte</td><td>An additional bit is added to make the number of 1s in the byte odd or even to match the parity. If a byte with an odd number of 1 bits is received when even parity is used, there is an error.</td></tr><tr><td>Parity block</td><td>Parity is calculated horizontally and vertically. A parity byte is created from the bits produced by the vertical parity check. This is sent with the data. The parity is re-checked when received and the position of an incorrect bit can be determined.</td></tr><tr><td>Checksum</td><td>A calculation is made from the data and the result transmitted with the data. The receiver repeats the calculation and compares the result with the value received. If the two are different, there is an error.</td></tr></table>	Method	Description	Parity byte	An additional bit is added to make the number of 1s in the byte odd or even to match the parity. If a byte with an odd number of 1 bits is received when even parity is used, there is an error.	Parity block	Parity is calculated horizontally and vertically. A parity byte is created from the bits produced by the vertical parity check. This is sent with the data. The parity is re-checked when received and the position of an incorrect bit can be determined.	Checksum	A calculation is made from the data and the result transmitted with the data. The receiver repeats the calculation and compares the result with the value received. If the two are different, there is an error.	4
Method	Description									
Parity byte	An additional bit is added to make the number of 1s in the byte odd or even to match the parity. If a byte with an odd number of 1 bits is received when even parity is used, there is an error.									
Parity block	Parity is calculated horizontally and vertically. A parity byte is created from the bits produced by the vertical parity check. This is sent with the data. The parity is re-checked when received and the position of an incorrect bit can be determined.									
Checksum	A calculation is made from the data and the result transmitted with the data. The receiver repeats the calculation and compares the result with the value received. If the two are different, there is an error.									
7(f)(ii)	1 mark each to max 2: • Encodes/scrambles data • ... so if it is intercepted it cannot be understood • Algorithm/key is required to decode the data	2								

17. O/N 23/P12/Q5/a,b,c

5(a)	1 mark for: Either - Ensuring data can only be accessed by / disclosed to authorised persons Or - Ensuring data cannot be accessed by / disclosed to unauthorised persons	1
5(b)	1 mark for each bullet point (max 1) - Ensuring the accuracy / completeness / consistency of data (during / after processing) - Ensuring the data is up to date	1
5(c)	1 mark for each bullet point. Phishing email (max 2) - The email pretends to be from an official body ... persuading individuals to disclose private information // by example such as bank details ... or requesting authentication by redirecting to an unofficial/unauthorised website // inviting a user to click a link Spyware (max 2) - Malware downloaded without the user's knowledge ... which secretly records the user's actions / keystrokes on the computer ... and sends logs of the actions to a third party	4

18. O/N 23/P13/Q3/a(iii),c

3(a)(iii)	1 mark each for firewall, encryption and passwords. Firewall: - Monitors incoming and outgoing traffic and rejects any traffic that does not meet the set rules Encryption: - Ensures that if data is intercepted / obtained it cannot be understood without the decryption key Passwords: - Ensures only users with the correct password can access the resources // prevents unauthorised access	3
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3(c)	1 mark for each bullet point. • The administrator completes a visual check / checks by eye • ...that the tutor identifier input matches the tutor identifier on the original document • Double entry check // the administrator (or a second person) enters the number a second time • ...and the system compares it with the first entry	4
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19. O/N 23/P13/Q8/a,b,c

8(a)	1 mark for each bullet point (max 2) • Validation // a validation method named or described • ...protects the data by ensuring that the data is reasonable / sensible and within specified bounds	2
8(b)	1 mark for difference 1 mark for similarity Difference: • Pharming is malicious code that redirects to a fake website. Phishing uses an email to prompt user action. • Pharming is automatic. Phishing requires user action. Similarity: • Both try to obtain financial or personal information • Both are a false representation of an official organisation, e.g. a bank • Both make use of fake websites	2
8(c)	1 mark for each bullet point (max 3). • Download programs from reputable websites / sources • ...as these are less likely to contain malware • Backup / archive computer systems • ...so they can be restored in case of data loss from malware program installation • Install and run anti-malware program • ...so that regular scans can be made for known malware • ...and if malware is found it can be quarantined / removed • ...and computer's anti-malware definitions are regularly updated • Using a firewall to block unused ports • ...so that malware cannot enter the computer system • Deny administrator privileges to everyday users • ...so that malware cannot be downloaded by everyday users • Avoid the use of / access to removable devices • ...so that malware cannot be installed from these devices	3

20. M/J 23/P12/Q2/c(i)

2(c)(i)	1 mark each to max 2 • Presence check to make sure that the (rider level) is entered • Look-up / Existence check to make sure the rider level is only Beginner, Intermediate or Advanced • Length check to make sure the rider level entered is either 8 or 12 characters • Type check to make sure the rider level is alphanumeric	2
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21. M/J 23/P13/Q4/d

4(d)(i)	1 mark each to max 2 • Length check: the registration number must be 6 characters long • Format check: the registration number must be in the format letter-digit-digit-digit-letter-letter • Type check: the registration number must be alphanumeric	2
4(d)(ii)	1 mark each • Visual check: Manually compare the registration number entered with the source document • Double entry: Enter the registration number twice and the computer compares to check they are the same	2
4(d)(iii)	The registration number on the original document might be in the correct format but may be the incorrect registration number for that car.	1

22. M/J 23/P13/Q6/a,b

6(a)	1 mark each to max 5 • The sender hashes the document • ... to produce a <u>digest</u> • The sender <u>encrypts</u> the digest to create the digital signature • The message and the signature are sent to the receiver • The receiver <u>decrypts</u> the signature to reproduce the digest • The receiver uses the <u>same</u> hashing algorithm on the document received to produce a second digest • The receiver compares this digest with the one from the digital signature • If both of the receiver's digests are the same the document is authentic	5
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6(b)	1 mark each for identification and appropriate description of 2 pieces of software, max 2 <table><tr><th>Type of software</th><th>Description</th></tr><tr><td>Antivirus</td><td> - scans the computer for viruses and checks against a stored database of viruses, that needs to be updated regularly and then deletes / quarantines them - compares downloaded files to a database of known viruses and prevents the download continuing </td></tr><tr><td>Antispyware</td><td> - scans the computer for spyware and checks against a stored database of viruses, that needs to be updated regularly and then deletes / quarantines them - compares downloaded files to a database of known spyware and prevents the download continuing. </td></tr><tr><td>Firewall</td><td> - monitors incoming and outgoing traffic and compares it to criteria that are set by the user such as through a whitelist/blacklist/identifying allowed / blocked IP addresses - compares incoming and outgoing traffic to criteria blocks those that do not match criteria </td></tr><tr><td>Antimalware</td><td> - scans the computer for viruses and checks against a stored database of viruses, that needs to be updated regularly and then deletes / quarantines them - compares downloaded files to a database of known viruses and prevents the download continuing </td></tr></table>	Type of software	Description	Antivirus	- scans the computer for viruses and checks against a stored database of viruses, that needs to be updated regularly and then deletes / quarantines them - compares downloaded files to a database of known viruses and prevents the download continuing	Antispyware	- scans the computer for spyware and checks against a stored database of viruses, that needs to be updated regularly and then deletes / quarantines them - compares downloaded files to a database of known spyware and prevents the download continuing.	Firewall	- monitors incoming and outgoing traffic and compares it to criteria that are set by the user such as through a whitelist/blacklist/identifying allowed / blocked IP addresses - compares incoming and outgoing traffic to criteria blocks those that do not match criteria	Antimalware	- scans the computer for viruses and checks against a stored database of viruses, that needs to be updated regularly and then deletes / quarantines them - compares downloaded files to a database of known viruses and prevents the download continuing	2
Type of software	Description											
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Antimalware	- scans the computer for viruses and checks against a stored database of viruses, that needs to be updated regularly and then deletes / quarantines them - compares downloaded files to a database of known viruses and prevents the download continuing											

23. M/J 23/P23/Q2/b

2(b)(i)	MP1 Value for <u>month</u> is between 1 and 12 (inclusive) MP2 Value of <u>year</u> is <= 2002	2
2(b)(ii)	MP1 Reference to <u>month</u> and <u>day</u> MP2 Clear description for a <u>check</u> that the day number matches with a relevant month (Either day matches with month // month matches with day)	2

24. O/N 22/P11/Q2

2	1 mark for each correct line. Security feature firewall pharming anti-virus software encryption converts data to an alternative form redirects a user to a false website verifies the authenticity of data scans files on the hard drive for malicious software accepts or rejects incoming and outgoing packets based on criteria	4
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25. O/N 22/P12/Q4/a,b,c

4(a)	data verification is checking if input data is the same as the original whereas data validation is checking that the data is reasonable / sensible	1
4(b)	1 mark for each bullet point (max 3): - checksum value is calculated from the data before transmission // correct description of a checksum algorithm ... this calculated value is transmitted with the data receiving computer recalculates the checksum from the received data - if the checksum received and calculated match, no error has occurred // if the checksum received and calculated do not match, an error has occurred	3
4(c)	1 mark for each bullet point (max 2). For example: - to make sure data is in the required format // only expected characters allowed - to make sure the data is already present in the system - to make sure the data contains the correct number of characters - to ensure that non-numeric data is entered	2

26. O/N 22/P12/Q6/a

6(a)(i)	1 mark for each method of ensuring authenticity (max 2): • (email) message put through hashing algorithm to produce a digest • Digest encrypted with <u>sender's private</u> key (to create the digital signature) • the (digital) signature can only be decrypted with matching <u>sender's public</u> key	2
6(a)(ii)	1 mark for each bullet point: • monitors <u>incoming</u> and <u>outgoing</u> packets / traffic • checks against an allow list / deny list of IP addresses // checks against a set of rules for acceptable data / ports etc. • blocks transmissions that do not meet criteria / rules // allows through is satisfies the criteria /rules	3

27. O/N 21/P11/Q2/a,b,c

2(a)	1 mark per bullet point • security is protecting data from loss / corruption • integrity is ensuring the consistency / accuracy of the data	2
2(b)(i)	1 mark per bullet point • validation checks that data is reasonable / sensible • example e.g. checking data is the right number / type of characters	2
2(b)(ii)	1 mark per bullet point • verification checks that data is the same as the original • by example e.g. double entry	2
2(c)	1 mark per similarity to max 2 • Both are pieces of malicious software • Both are downloaded / installed/run without the user's knowledge • Both can pretend to be / are embedded in other legitimate software when downloaded // both try to avoid the firewall • Both run in the background 1 mark for difference • Virus can damage computer data; spyware only records / accesses data • Virus does not send data out of the computer; spyware sends recorded data to third party • Virus replicates itself; spyware does not replicate itself	3

28. O/N 21/P12/Q1

1	1 mark for 3 correct lines only from Data Security 1 mark for 2 correct lines only from Data Integrity <pre> graph LR Firewall --> DS[Data Security] DoubleEntry[Double entry] --> DS PresenceCheck[Presence check] --> DI[Data Integrity] AccessRights[Access rights] --> DI Password --> DS </pre>	2
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29. O/N 21/P13/Q2/a,b,c

2(a)	1 mark per bullet point • security is protecting data from loss / corruption • integrity is ensuring the consistency / accuracy of the data	2
2(b)(i)	1 mark per bullet point • validation checks that data is reasonable / sensible • example e.g. checking data is the right number / type of characters	2
2(b)(ii)	1 mark per bullet point • verification checks that data is the same as the original • by example e.g. double entry	2
2(c)	1 mark per similarity to max 2 • Both are pieces of malicious software • Both are downloaded / installed/run without the user's knowledge • Both can pretend to be / are embedded in other legitimate software when downloaded // both try to avoid the firewall • Both run in the background 1 mark for difference • Virus can damage computer data; spyware only records / accesses data • Virus does not send data out of the computer; spyware sends recorded data to third party • Virus replicates itself; spyware does not replicate itself	3

30. M/J 21/P11/Q6/a,b,c

6(a)	Range (check)	1
6(b)	Presence (check)	1
6(c)	Existence (check)	1

31. M/J 21/P12/Q8/a,b,c

8(a)	1 mark per bullet point • Security protects data against loss • Privacy protects data against unauthorised access	2
8(b)	1 mark for a correct answer • Two factor authentication • Biometric passwords • Key Card Access • Firewall	1
8(c)	1 mark per correct answer to max 2 • Malware // viruses // spyware // by example • Hacking • Phishing • Pharming	2

32. M/J 21/P13/Q6/a,b,c

6(a)	Range (check)	1
6(b)	Presence (check)	1
6(c)	Existence (check)	1